

①

Math test - 3

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Section - AQuestion - 1

$$|A| = -2$$

$$|kA| = k^n |A|$$

$$|-3A| = (-3)^3 |A| = -27 \times (-2)$$

$$= 54$$

Question - 2

$$\frac{1}{2} \times \frac{4}{5} + \frac{1}{2} \times \frac{2}{5} = \frac{4}{10} + \frac{2}{10} = \frac{6}{10} = \frac{3}{5}$$

Question - 3

$$[0, \pi]$$

Question - 4

$$\vec{a} \cdot \vec{b} = ?$$

$$\vec{a} = \hat{i} - 2\hat{j} + \hat{k} \quad \vec{b} = 3\hat{i} + \hat{j} - 2\hat{k}$$

$$\vec{a} \cdot \vec{b} = (1 \times 3) + (-2 \times 1) + (1 \times -2)$$

$$= 3 - 2 - 2 = 3 - 4 = -1$$

Question - 5

$$P(B/A) = \frac{P(B \cap A)}{P(A)}, \text{ where } P(A) \neq 0$$

Section - B

Question - 6

$$R = \{(a, b) : a \leq b^2\}$$

$$\text{Let } a = 2, b = -2, c = 1$$

$$(a, b) \in R \text{ as } 2 \leq (-2)^2 \Rightarrow 2 \leq 4, \text{ true}$$

$$(b, c) \in R \text{ as } -2 \leq 1^2 \Rightarrow -2 \leq 1, \text{ true}$$

$$(a, c) \notin R \text{ as } 2 \not\leq 1^2 \Rightarrow 2 \not\leq 1, \text{ false}$$

∴ R is not transitive.

Question - 7

$$y = \sin^{-1}(2x\sqrt{1-x^2})$$

$$\text{Let } x = \sin \theta$$

$$\theta = \sin^{-1}(x)$$

$$y = \sin^{-1}(2\sin \theta (\sqrt{1-\sin^2 \theta}))$$

$$= \sin^{-1}(2\sin \theta \cos \theta)$$

$$= \sin^{-1}(\sin(2\theta))$$

$$= 2\theta$$

$$y = 2\sin^{-1}(x)$$

After differentiating,

$$\boxed{\frac{dy}{dx} = \frac{2}{\sqrt{1-x^2}}}$$

Question - 8

$$\vec{a} = \hat{i} + \hat{j} \quad \vec{b} = \hat{j} + \hat{k}$$

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = \hat{i}(1-0) - \hat{j}(1-0) + \hat{k}(1-0) \\ = \hat{i} - \hat{j} + \hat{k}$$

$$|\vec{a} \times \vec{b}| = \sqrt{1^2 + (-1)^2 + 1^2} = \sqrt{3}$$

Unit vector, $\hat{u} = \pm \frac{\vec{a} \times \vec{b}}{|\vec{a} \times \vec{b}|} = \pm \frac{\hat{i} - \hat{j} + \hat{k}}{\sqrt{3}}$

Question - 9

$$\Delta = \begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \\ 6x & 9x & 12x \end{vmatrix}$$

$$\Delta = 3x \begin{vmatrix} 2 & 3 & 4 \\ 5 & 6 & 8 \\ 2 & 3 & 4 \end{vmatrix}$$

Since R_1 & R_3 are equal,

$$\Delta = 3x \times 0 = 0$$

Question - 10

$$A = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \quad B = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$$

$$(A^T + B)^2 = A^2 + B^2$$

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~~RHS~~

$$(A+B)^2 = \left(\begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} + \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \right)^2$$

$$= \begin{bmatrix} 1+a & 0 \\ 2+b & -2 \end{bmatrix}^2$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$(A+B)^2 = A^2 + B^2$$

$$A^2 + B^2 + AB + BA = A^2 + B^2$$

$$AB + BA = 0$$

$$AB + BA = 0$$

$$AB = \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix}$$

$$= \begin{bmatrix} a-b & 1+1 \\ 2a-b & 2+1 \end{bmatrix} = \begin{bmatrix} a-b & 2 \\ 2a-b & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} a & 1 \\ b & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} a+2 & -a-1 \\ b-2 & -b+1 \end{bmatrix} = \begin{bmatrix} a+2 & -(a+1) \\ b-2 & 1-b \end{bmatrix}$$

$$\begin{bmatrix} a-b & 2 \\ 2a-b & 3 \end{bmatrix} = - \begin{bmatrix} a+2 & -a-1 \\ b-2 & 1-b \end{bmatrix}$$

$$\begin{bmatrix} a-b & 2 \\ 2a-b & 3 \end{bmatrix} = \begin{bmatrix} -a-2 & a+1 \\ 2-b & b-1 \end{bmatrix}$$

$$a+1=2$$

$$\boxed{a=1}$$

$$b-1=3$$

$$\boxed{b=4}$$

$$a=1, b=4$$

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Question-11

$$f(x) = \begin{cases} \frac{x^2-4}{x-2}, & \text{if } x \neq 2 \\ 4, & \text{if } x = 2 \end{cases}$$

$$x = 2$$

LHL

$$\begin{aligned} \lim_{x \rightarrow 2^-} f(x) &= \lim_{x \rightarrow 2^-} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2^-} \frac{(x-2)(x+2)}{(x-2)} \\ &= \lim_{x \rightarrow 2^-} (x+2) = 2+2 = 4. \end{aligned}$$

RHL

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{x^2-4}{x-2} = \lim_{x \rightarrow 2^+} (x+2) = 2+2 = 4$$

Value at $x=2$,

$$f(2) = 4$$

Since $\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x) = f(2) = 4$, the

function is continuous at $x=2$.

Question-12

$$f(x) = 2x^3 - 3x^2 - 36x + 7$$

$$f'(x) = 6x^2 - 6x - 36 = 6(x^2 - x - 6) = 6(x-3)(x+2)$$

Critical points, $x = -2, x = 3$

Intervals:

$$\text{For } x \in (-\infty, -2) : f'(x) > 0$$

$$\text{For } x \in (-2, 3) : f'(x) < 0$$

$$\text{For } x \in (3, \infty) : f'(x) > 0$$

Increasing: $(-\infty, -2)$ and $(3, \infty)$

Decreasing: $(-2, 3)$

Question-13

$$y = e^{a \cos^{-1}(x)}$$

Differentiating w.r.t x

$$y_1 = \frac{dy}{dx} = e^{a \cos^{-1} x} \cdot \left(\frac{-a}{\sqrt{1-x^2}} \right) = \frac{-ay}{\sqrt{1-x^2}}$$

Square both sides.

$$(1-x^2) \cdot y_1^2 = a^2 y^2$$

Differentiating:

$$(1-x^2) \cdot 2y_1 y_2 + (y_1^2) (-2x) = a^2 \cdot 2yy_1$$

Divide throughout by $2y_1$

$$(1-x^2) y_2 - x y_1^2 = a^2 y$$

$$\Rightarrow \left[(1-x^2) \frac{d^2 y}{dx^2} - x \left(\frac{dy}{dx} \right)^2 - a^2 y = 0 \right]$$

Question-14

$$\tan^{-1}\left(\frac{1}{4}\right) + \tan^{-1}\left(\frac{2}{9}\right) = \frac{1}{2} \cos^{-1}\left(\frac{3}{5}\right)$$

LHS

$$\tan^{-1}\left(\frac{\frac{9+8}{36}}{1 - \frac{1}{4} \times \frac{2}{9}}\right) = \tan^{-1}\left(\frac{\frac{17}{36}}{\frac{34}{36}}\right) = \tan^{-1}\left(\frac{1}{2}\right)$$

$$\tan^{-1}\left(\frac{1}{2}\right) = \frac{1}{2} \left(2 \tan^{-1}\left(\frac{1}{2}\right) \right)$$

$$2 \tan^{-1} x = \cos^{-1} \left(\frac{1-x^2}{1+x^2} \right)$$

$$2 \tan^{-1} \left(\frac{1}{2} \right) = \cos^{-1} \left(\frac{1 - \frac{1}{4}}{1 + \frac{1}{4}} \right) = \cos^{-1} \left(\frac{3/4}{5/4} \right)$$

$$= \cos^{-1} \left(\frac{3}{5} \right)$$

∴ LHS = RHS. //

Question - 15

Let E_1 = die lands on 6

$$P(E_1) = \frac{1}{6}$$

Let E_2 = die doesn't land on 6

$$P(E_2) = \frac{5}{6}$$

Let A = The man reports it is 6

$$P(A/E_1) = \frac{3}{4} \quad P(A/E_2) = \frac{1}{4}$$

By Baye's theorem:

$$P(E_1/A) = \frac{P(E_1) \cdot P(A/E_1)}{P(E_1) \cdot P(A/E_1) + P(E_2) \cdot P(A/E_2)}$$

$$= \frac{\frac{1}{6} \times \frac{3}{4}}{\frac{1}{6} \times \frac{3}{4} + \frac{5}{6} \times \frac{1}{4}} = \frac{\frac{3}{24}}{\frac{3+5}{24}}$$

$$\boxed{P(E_1/A) = \frac{3}{8}} //$$

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Section - D

Question - 16

$$A = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A - 4I = 0$$

$$A^3 = A^2 \cdot A$$

$$A^2 = \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4+1+1 & -2-2-1 & 2+1+2 \\ -2-2-1 & 1+4+1 & -1-2-2 \\ 2+1+2 & -1-2-2 & 1+1+4 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 12+5+5 & -6-10-5 & 6+5+10 \\ -10-6-5 & 5+12+5 & -5+6-10 \\ 10+5+6 & -5-10-6 & 5+5+12 \end{bmatrix}$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix}$$

$$6A^2 = 6 \begin{bmatrix} 6 & -5 & 5 \\ -5 & 6 & -5 \\ 5 & -5 & 6 \end{bmatrix} = \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix}$$

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$$9A = 9 \begin{bmatrix} 2 & -1 & 1 \\ -1 & 2 & -1 \\ 1 & -1 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix}$$

$$4I = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$A^3 - 6A^2 + 9A - 4I$$

$$= \begin{bmatrix} 22 & -21 & 21 \\ -21 & 22 & -21 \\ 21 & -21 & 22 \end{bmatrix} - \begin{bmatrix} 36 & -30 & 30 \\ -30 & 36 & -30 \\ 30 & -30 & 36 \end{bmatrix} + \begin{bmatrix} 18 & -9 & 9 \\ -9 & 18 & -9 \\ 9 & -9 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O$$

∴ Hence proved.

$$A^{-1} = \frac{\text{adj } A}{|A|}$$

$$|A| = 2(4+1) + 1(-2+1) + 1(1-2)$$

$$= 2 \times 5 + 1 \times -1 + 1 \times -1$$

$$= 10 - 1 - 1$$

$$|A| = 8$$

$$\text{adj } A = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 5 \end{bmatrix}^T = \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 5 \end{bmatrix}$$

$$A^{-1} = \frac{1}{8} \begin{bmatrix} 5 & 1 & -1 \\ 1 & 3 & 1 \\ -1 & 1 & 5 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} 5/8 & 1/8 & -1/8 \\ 1/8 & 3/8 & 1/8 \\ -1/8 & 1/8 & 5/8 \end{bmatrix}$$

(10)

(P)

Question - 17

$$A = \{1, 2, 3, 4, 5\}$$

$$R = \{(a, b) : |a - b| \text{ is even}\}$$

ReflexiveFor any $a \in A$,

$$|a - a| = 0 = 2(0)$$

Since 0 is an integer, $(a, a) \in R$ for all $a \in A$. $\therefore R$ is reflexiveSymmetric

$$\text{Let } (a, b) \in R$$

$$\Rightarrow |a - b| = 2k, \quad k \in \mathbb{Z}, \quad 2 \neq 0$$

$$\text{Since } |b - a| = |-(a - b)| = |a - b|$$

$$|b - a| = 2k$$

$$(b, a) \in R, \quad \therefore R \text{ is symmetric}$$

Transitive

$$\text{Let } (a, b) \in R \text{ \& } (b, c) \in R$$

$$|a - b| \text{ is even \& } |b - c| \text{ is even}$$

$$\Rightarrow a - b = 2m \text{ \& } b - c = 2n, \quad m, n \in \mathbb{Z}$$

Add the 2 equations

$$a - b + b - c = 2m + 2n \Rightarrow a - c = 2(m + n)$$

Since $m + n \in \mathbb{Z}$, $(a - c)$ is an even integer

$$\Rightarrow |a - c| \text{ is even}$$

$$\Rightarrow (a, c) \in R$$

 $\therefore R$ is transitive $\therefore$ Since R is reflexive, symmetric, \& transitive, it is an equivalence relation.